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DEVONIAN PALEOMAGNETISM OF YUNNAN PROVINCE ACROSS THE SHAN THAI-SOUTH CHINA SUTURE

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Abstract. A paleomagnetic study of Devonian sedimentary sequences from western and eastern Yunnan, China, yields contrasting paleolatitudes for these two areas. They indicate an equatorial position for eastern Yunnan, as part of the Yangtze Paraplatform or South China Block, and a paleolatitude of about 40° for western Yunnan as part of the Shan State (or Shan-Thai) Block, which probably was part of the Gondwana supercontinent. The majority of samples is strongly overprinted by a present-day field magnetization, but thermal demagnetization has been successful in isolating ancient, dual-polarity characteristic directions of magnetization in rocks from eastern and western Yunnan, with results from the latter passing the fold test. The paleopole from eastern Yunnan is located at 8.9°N, 10.4°E (declination/inclination = 279.2°/+2.7°, k = 13, alpha 95 = 6.9°, N = 33 samples from 7 sites). The paleopole from western Yunnan is located at 66.5°N, 133.8°E (declination/inclination = 18.0°/+62.0°, k = 21, alpha 95 = 5.8°, N = 31 samples from 7 sites), yielding a paleolatitude of 42°. The two blocks studied are separated by the Red (Yuan) River and Lancang (Mekong) River fault zones, one of which is inferred to be an ancient suture.

INTRODUCTION

Following a very early mobilistic synthesis of Argand [1924], it has repeatedly been argued that Southeast Asia is a mosaic of continental blocks which coalesced at various times between the Late Paleozoic and the Middle Tertiary [e.g., Whittington and Hughes, 1972; Burrett, 1974; Haile, 1981; Mitchell, 1981; Stauffer, 1983]. This paper concerns itself with two adjacent blocks: the South China Block or Yangtze Paraplatform and the Shan Thai Block of Burma, Thailand, and western Yunnan, China (Figure 1).

The South China Block is one of the major crustal blocks of Asia; it has been the subject of several paleomagnetic studies although its Phanerozoic Apparent Polar Wander Path (APWP) is still incompletely known [e.g., Chan et al., 1984; Fang et al., 1988-89; Huang et al., 1986; Kent et al., 1986, 1987; Lin et al., 1985; McElhinny et al., 1981; Opdyke et al., 1986, 1987; Zhao, 1987]. Particularly for pre-Permian times, the data base is very sparse; this stems in part from the problems related to strong present-day geomagnetic field overprinting [Kent et al., 1987; Zhao, 1987].

All available pre-Permian paleomagnetic directions from the South China Block have shallow inclinations, uniformly indicating equatorial paleolatitudes (e.g., Opdyke et al. [1987] for the Silurian, Lin et al. [1985] for the Carboniferous and Early Cambrian, and Fang et al. [1988-89] for the Precambrian-Cambrian boundary). Liu et al. [1986] also collected samples (38 total) from pre-Permian rocks (Cambrian, Silurian, and Devonian from Qujing, eastern Yunnan) and reported shallow inclinations as well, although they did not publish details. The latter study observed northeasterly declinations, whereas the Silurian and Early Cambrian rocks of the other studies revealed easterly declinations; since some of these differences in declination are obtained from the same Silurian rocks (e.g., Liu et al. [1986] versus Opdyke et al. [1987]), this could be taken to suggest that demagnetizations were not complete, remagnetizations in some of the rocks occurred in younger Paleozoic time, or the APWP is very complex. The (Late Precambrian-) Early Cambrian declinations are equally inconsistent: Fang et al. [1988-89] obtained northerly declinations, whereas Lin et al. [1985] obtained east-west declinations. It is clear therefore that the Paleozoic APWP segment for the South China Block must be regarded with circumspection at this time. Our experience with the paleomagnetism of North American Paleozoic

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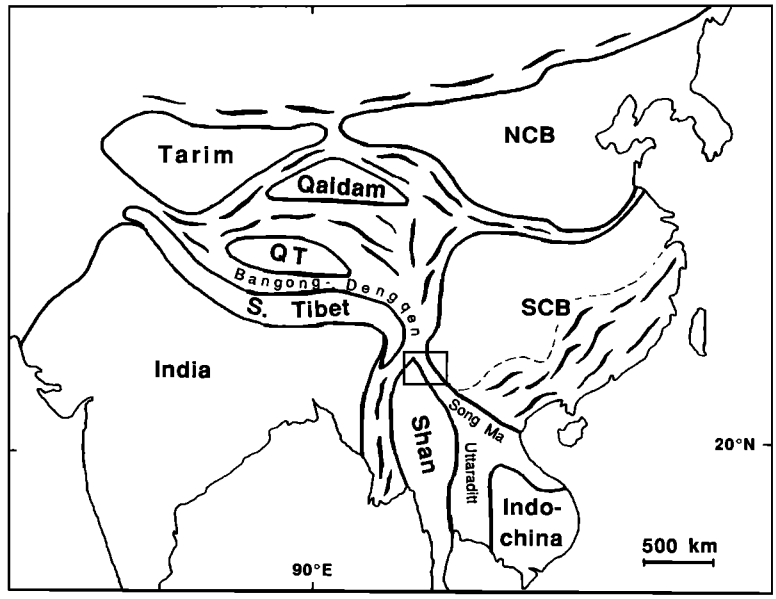


Fig. 1. Sketch map of the tectonic elements of southeastern Asia, modified from Li et al. [1982] and Achache and Courtillot [1985]. The small rectangle outlines the sampling area of Figure 2. QT, Qiangtang Block; Shan, Shan State Massif. The Bagong-Dengqen, Song Ma, and Uttaraditt suture zones are indicated.

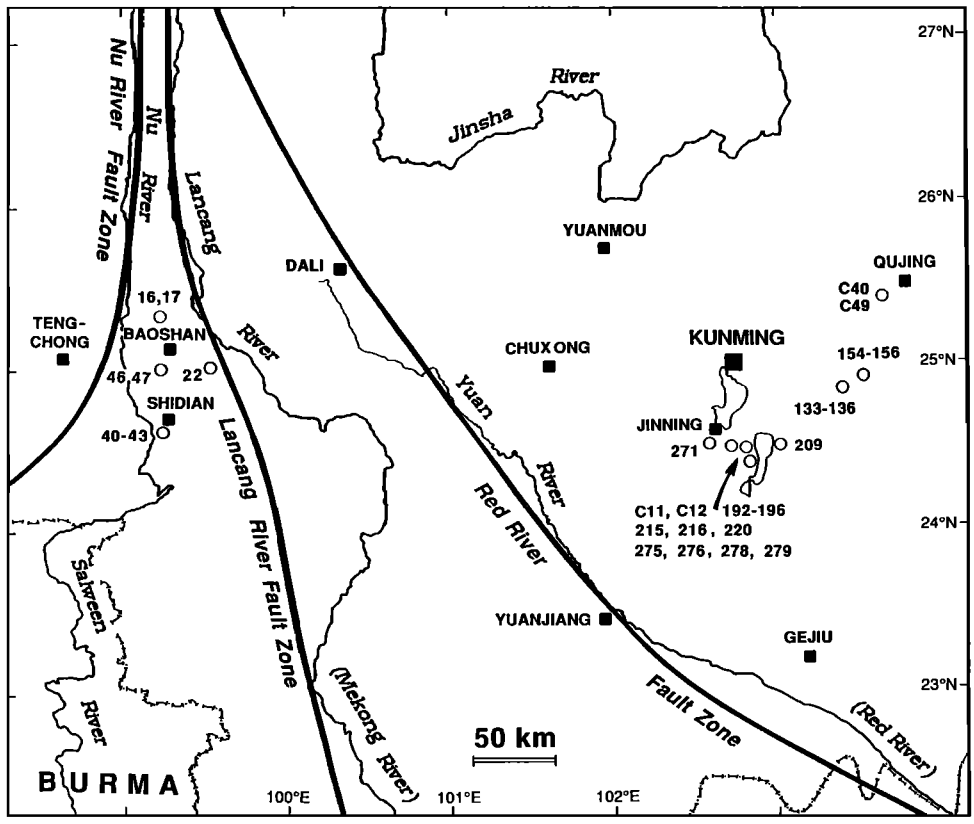


Fig. 2. Regional map of eastern and western Yunnan, with sampling sites (open circles) indicated.

Fig. 3. Devonian stratigraphic units in Yunnan Province.

Series	eastern Yunnan	western Yunnan
Upper Devonian	Zaige Formation (6 sites) Grey colored massive limestone. 460 m	Dazaimen Formation (1 site) Dark grey argillaceous limestone.
Middle Devonian	Haikou Formation (9 sites) Yellow and green sandstones, quartzites and shales. Contains: *Bothriolepis sinensis, Qunasipetalichthys hakouensis, etc..* 15-100 m	Heyuanzai Formation (3 sites) Grey , red and yellow brecciated limestone.
	Qingmen Formation (8 sites) Light colored fine limestones. 40 m	
Lower Devonian	Cuifengshan Formation (2 sites) Yellow and green thick-bedded sandstones with intercalated shales. Contains: *Galeaspis xiujiachongensis, Yunnanplepis , Polybranchiaspis, etc..* 1500 m	Wenquan Formation (5 sites) Dark grey thin-bedded siliceous limestone, yellow-red argillite and argillaceous limestone.

Fig. 3. Devonian stratigraphic units in eastern and western Yunnan [from Liang et al., 1984; Yang et al., 1986]. The number of sites collected is indicated.

platform carbonates is that a large majority of those rocks are remagnetized in later Paleozoic times [e.g., Miller and Kent, 1988; Van der Voo, 1989]; a similar scenario for the Yangtze Paraplatform cannot be precluded.

Nevertheless, the Yangtze Paraplatform contains many exposures of paleomagnetically suitable, unmetamorphosed, Paleozoic sequences ranging in age from Late Precambrian to Permian; moreover, the middle Mesozoic Indosinian (Yanshanian?) orogeny, which had a profound effect in southeastern China [Yang et al., 1986], hardly affected the rocks in eastern Yunnan. As a cooperative project of the Geological Institute of Yunnan Province and the University of Michigan, we sampled the Devonian strata in eastern Yunnan. Our sampling was extended across the Lancang River into western Yunnan (Figures 1 and 2), which is thought to form the northern end of the Thailand-Burma Shan State Massif, in the hopes of finding contrasting paleomagnetic signatures between eastern and western Yunnan in order to document relative displacements between these southeast Asian fragments.

GEOLOGY AND SAMPLING

Eastern Yunnan Province forms part of the South China Block, which for about 600 m.y. (Late Precambrian through Mesozoic) remained a stable platform experiencing only minor vertical movements. While a few regional disconformities occur, the section of shallow marine and epicontinental sequences is nearly complete. Folding took place in the middle Tertiary [Liang et al., 1984], resulting in north-south trending fold axes in the province.

The Devonian System in eastern Yunnan is best exposed in the area around Kunming, Jinning, and Qujing (Figure 2; 24.5°N, 103.0°E). Upper Silurian-lowermost Devonian thin-bedded limestones of the Yulongsi Formation [Lin, 1980] conformably underlie the Devonian, which is followed disconformably by Lower Carboniferous limestones of the Datang Formation. The Devonian sequence is divided into three series (Figure 3), each well dated by fossils. From this sequence we have collected a total of 156 samples from 25 sites (Figure 2). High-angle faults in the sampling areas disect a gently undulating set of folds in the pre-Tertiary strata; dips on the flanks of these gentle folds generally range between 5° and 35° in different directions (see Table 1).

Located about 200 to 400 km to the west of Kunming, several very large NNW trending fracture zones form the boundary of the South China Block: the Nu River, Lancang (Mekong) River, and Red (Yuan) River fault zones. The area near Baoshan (Figure 2), between the Nu River and Lancang River fault zones, is thought to represent the northern end of the Shan State Massif, which is part of a tectonic element (Shan-Thai Block) in Burma and western Thailand that links southern Tibet with the Malay Peninsula. It has been proposed that the Shan-Thai Block was part of Gondwana until the Late Paleozoic [Ridd, 1971; Mitchell, 1981]. Li et al. [1982] and Chen et al. [1987] further proposed that the Lancang River

TABLE 1. Paleomagnetic Results From Eastern Yunnan

Site	Strike/Dip	N_{pdf}	N_{random} unstable	N_{char}	N_o	pdf Direction			Characteristic Direction				
						$D/I_{in\ situ}$	k	α_{95}	$D/I_{in\ situ}$	$D/I_{tilt\ corr}$	k	α_{95}	Range
Upper Devonian Zaige Formation													
193*	227/19	1	1	4	6	352.3/44.9	-	-	319.1/20.6	318.7/ 1.8	100.1	9.2	100-410
194	248/20	6	0	0	6	11.1/45.7	472.0	3.1	- -	- -	-	-	100-410
195	226/15	7	0	0	7	357.5/43.4	39.9	9.7	- -	- -	-	-	200-410
278	320/16	3	0	0	3	6.4/47.3	59.4	16.1	- -	- -	-	-	5-85
279	315/13	4	0	0	4	9.4/41.4	87.1	9.9	- -	- -	-	-	5-85
C12	19/35	9	0	0	9	357.8/42.9	130.3	4.5	- -	- -	-	-	100-410
Middle Devonian Haikou Formation													
154*	74/21	0	4	1	5	- -	-	-	259.6/ 9.3	255.9/10.6	-	-	100-650
155	12/15	0	4	0	4	- -	-	-	- -	- -	-	-	100-410
156	294/39	0	8	0	8	- -	-	-	- -	- -	-	-	100-450
192*	244/20	0	0	9	9	- -	-	-	261.7/ 7.8	268.1/-0.2	18.5	11.3	300-550
196*	48/34	0	5	3	8	- -	-	-	269.7/-15.8	271.4/7.7	20.9	20.5	100-500
271	84/16	7	0	0	7	2.7/46.4	43.0	9.3	- -	- -	-	-	100-350
275*	276/34	0	0	3	3	- -	-	-	305.2/ 5.0	303.5/-11.5	29.2	23.3	200-620
276	306/15	0	0	(6)	6	- -	-	-	(35.1/13.9)	(34.4/-2.4)	19.2	15.7	200-620
C11*	324/25	0	4	8	12	- -	-	-	95.5/ 9.7	95.4/-9.2	174.6	4.2	150-590
Middle Devonian Qingmen Formation													
133	6/24	4	0	0	4	350.3/41.0	170.9	7.0	- -	- -	-	-	250-510
134	25/14	6	0	0	6	352.1/48.3	97.8	6.8	- -	- -	-	-	200-500
135	359/13	7	0	0	7	350.8/47.8	43.9	9.2	- -	- -	-	-	200-450
136	359/13	6	0	0	6	2.7/40.2	87.8	7.2	- -	- -	-	-	200-410
209	272/31	3	2	0	5	356.1/40.1	71.1	7.5	- -	- -	-	-	100-410
215	353/8	5	0	0	5	4.2/39.6	77.9	8.7	- -	- -	-	-	100-410
216	9/23	1	3	0	4	9.9/50.2	-	-	- -	- -	-	-	100-410
220	239/25	6	0	0	6	350.6/47.9	74.9	7.8	- -	- -	-	-	100-620
Lower Devonian Cuifengshan Formation													
C40*	332/14	1	2	5	8	350.2/40.9	-	-	280.2/-13.2	278.6/-2.0	22.6	16.4	100-400
C49	304/33	2	6	0	8	345.0/43.5	-	-	- -	- -	-	-	100-500

Strike/Dip indicates bedding orientation; N_{pdf}, N_{random}, $N_{unstable}$, and N_{char} indicate number of samples which revealed present day field, random, unstable, and characteristic directions; N_o indicates number of demagnetized samples; pdf, present day field; char., characteristic component; D/I, declination/inclination; k, precision parameter; α_{95}, radius of 95 percent of confidence; Range, blocking temperature in °C or AF field in mT.

Symbol * indicates characteristic directions used to calculate means.

fault zone, connecting the Bangong-Dengqen suture zone in Tibet (Figure 1) with the Gulf of Thailand, is a major boundary between Gondwana-derived elements and Cathaysian-Eurasian elements and thus represents the zone in which the Tethys disappeared; to the southeast, this zone branches into the Uttaraditt suture between the Shan-Thai Block and the Indochina Block and the Song Ma suture between Indochina and the South China Block [e.g., Achache and Courtillot, 1985; Tapponnier et al., 1986; Peltzer and Tapponnier, 1988]. The Devonian System of western Yunnan is fossiliferous and, like that of the South China Block, divided into three series (Figure 3). The rocks consist of thin-bedded platform carbonates near Baoshan and Shidian (25.0°N, 99.0°E), and a total of 48 samples from 9 sites were collected (Figure 2). Strikes in this area are NNW-SSE and dips range up to 60° (see Table 2); many high-angle faults have been mapped in this area, and folding has clearly been more intense than in Eastern Yunnan.

A gasoline-powered portable drill was used to extract samples in the field, and these were oriented with a Brunton compass. In the laboratory the 2.5 cm diameter samples were cut into specimens of 2.2 cm height.

PALEOMAGNETIC RESULTS

Measurements of the Natural Remanent Magnetizations (NRM) were carried out on a ScT cryogenic magnetometer, with alternating field (AF) and thermal demagnetizations performed by Schonstedt equipment at the University of Michigan, in a shielded room with rest field generally less than 200 nT. Demagnetization results have been plotted in orthogonal vector diagrams [Zijderveld, 1967] and in stereographic projection, followed by Principal Component Analysis [Kirschvink, 1980]. The NRM intensities of the samples ranged between 0.2 and 5 mA/m, with no apparent correlation between intensities and the rock types as listed in Figure 3. The NRM directions clustered generally about the present-day geomagnetic field direction in Yunnan. With the exception of a few samples, AF demagnetization did not succeed in reducing the intensity to a small percentage of the initial NRM, so stepwise thermal demagnetization has been used on the majority of samples. The results are described below for eastern and western Yunnan and have been summarized in Tables 1, 2, and 3.

Eastern Yunnan

Four Devonian formations have been studied (Figure 3). Their results will be described sequentially from young to old.

The Upper Devonian Zaige Formation. AF and thermal demagnetization were equally effective in breaking down the NRM, but only one site yielded directions other than those conforming to the local present-day geomagnetic field (pdf). The pdf directions of the other five sites decayed univectorially to the origin of the demagnetization diagram, very much like the sample from another formation illustrated in Figure 4a. Table 1 gives site-mean directions for this pdf component as well as for the characteristic directions observed in site 193 of the Zaige Formation. The structural corrections for this

TABLE 2. Paleomagnetic Results From Western Yunnan

Site	Strike/Dip	N_{pdf}	N_{random} unstable	N_{char}	N_o	pdf Direction				Characteristic Direction				
						$D/I_{in\ situ}$	k	α_{95}		$D/I_{in\ situ}$	$D/I_{tilt\ corr}$	k	α_{95}	Range
Upper Devonian Dazaimen Formation														
17*	299/21	0	0	4	4	-	-	-	-	115.6/-70.4	162.3/-61.9	14.7	24.8	400-590
Middle Devonian Heyuanzai Formation														
16*	329/52	3	1	4	8	355.9/35.5	122.4	11.2		63.5/-56.4	231.2/-71.3	147.1	7.6	350-450
46*	126/61	0	2	3	5	-	-	-		195.3/ 0.7	178.7/-54.3	50.4	17.5	100-350
47*	137/27	0	0	5	5	-	-	-		215.5/-27.1	209.8/-53.3	170.2	5.9	300-550
Lower Devonian Wenquan Formation														
22	324/57	4	0	0	4	4.7/47.8	316.4	5.2		-	-	-	-	200-350
40*	201/37	0	0	7	7	-	-	-		55.3/56.0	352.9/58.4	59.8	7.9	350-690
41*	309/28	0	2	4	6	-	-	-		335.0/86.3	31.8/59.6	116.0	8.6	100-690
42	331/40	1	3	0	4	344.1/47.0	-	-		-	-	-	-	200-400
43*	346/40	0	1	4	5	-	-	-		307.8/77.4	57.8/56.6	31.4	16.7	200-500

See Table 1 for explanation.

TABLE 3. Formation Means

n/N	Declination/Inclination	k	α_{95}	Pole Position	
		Eastern Yunnan			
Upper Devonian Zaige Formation					
4/35	319.1/ 20.6	100.1	9.2	34.8°N	0.7°E
After TC	318.7/ 0.8	100.1	9.2	43.7°N	348.7°E
Middle Devonian Haikou Formation					
24/62	274.5/ -1.8	15.0	7.5	3.7°N	10.3°E
After TC	273.8/ 3.2	18.1	6.8	4.1°N	12.9°E
Lower Devonian Cuifengshan Formation					
5/16	280.2/-13.2	22.6	16.4	6.5°N	2.6°E
After TC	278.6/ -2.0	22.6	16.4	7.4°N	8.5°E
Lower and Middle Devonian mean					
29/78	275.4/ -3.6	16.6	6.8	4.2°N	9.1°E
After TC	274.5/ 2.4	19.0	6.1	4.6°N	12.2°E
Overall Averages for the Devonian of Eastern Yunnan					
33/113 (samples)	280.1/ -0.6	10.9	7.7	9.1°N	8.5°E
After TC	279.2/ 2.7	13.0	6.9	8.9°N	10.4°E
7/25 (site means)	281.0/ 0.6	10.0	20.1	10.1°N	8.7°E
After TC	281.4/ 2.4	12.6	17.7	10.9°N	9.3°E
6/7 (all R polarity)					
After TC	282.4/ 1.2	10.9	21.3	11.5°N	8.3°E
1/7 (N polarity)					
After TC	95.4/ -9.2	-	-	6.8°N	15.0°E
		Western Yunnan			
Upper Devonian Dazaimen Formation					
4/4	115.6/-70.4	14.7	24.8	34.8°S	239.6°E
After TC	162.3/-61.9	14.7	24.8	66.8°S	244.3°E
Middle Devonian Heyuanzai Formation					
12/18	215.5/ 6.3	3.9	25.4	45.5°S	43.4°E
After TC	205.2/-61.0	30.6	8.0	63.1°S	323.6°E
Lower Devonian Wenquan Formation					
15/26	39.0/74.3	14.2	10.5	45.5°N	125.3°E
After TC	20.7/61.5	20.6	8.6	65.5°N	138.0°E
Lower and Middle Devonian mean					
27/44	39.1/79.6	3.4	17.7	39.7°N	115.6°E
After TC	22.8/61.3	24.7	5.7	64.4°N	140.7°E
Overall Averages for the Devonian of Western Yunnan					
31/48 (samples)	16.3/67.5	4.6	13.6	61.7°N	121.4°E
After TC	18.0/62.0	21.0	5.8	66.5°N	133.8°E
7/9 (site means)	6.6/67.2	3.9	35.4	64.5°N	108.9°E
After TC	19.5/62.1	27.5	11.7	65.7°N	135.1°E
3/7 (all R polarity)					
After TC	28.2/61.0	23.2	26.2	61.2°N	145.7°E
4/7 (all N polarity)					
After TC	192.6/-62.5	25.6	18.5	68.5°N	124.5°E

n/N, number of the samples used to calculate the means / number of the samples demagnetized; TC, tilt correction for bedding.

formation allow a fold test, and the pdf directions predictably fail the test (see also Figure 5), with k decreasing from 259 to 23 upon unfolding. The unblocking temperatures (100° - 410°C) and coercivities (5 - 85 mT) suggest that the remanence is carried by magnetite.

Four samples of site 193 revealed a northwesterly and shallow magnetization (Figure 6a), which is quite different from the pdf direction (both before and after tilt correction), with mean tilt-corrected declination/inclination (D/I) = 319°/+2° (Table 1).

The Middle Devonian Haikou Formation. Five of the total of nine sites of this formation revealed characteristic east-west shallow directions, one site had aberrant northeasterly shallow directions, one site showed only the pdf direction, and two sites displayed only unstable or random directions (Table 1). The characteristic direction is best revealed by thermal demagnetization (Figures 6b and 6c) and shows dual polarities, with a formation mean direction of D/I = 274°/+3° after tilt correction (N = 24 samples from five sites, k = 18, and alpha 95 = 6.8° ; see Table 3). Acquisition of isothermal remanent

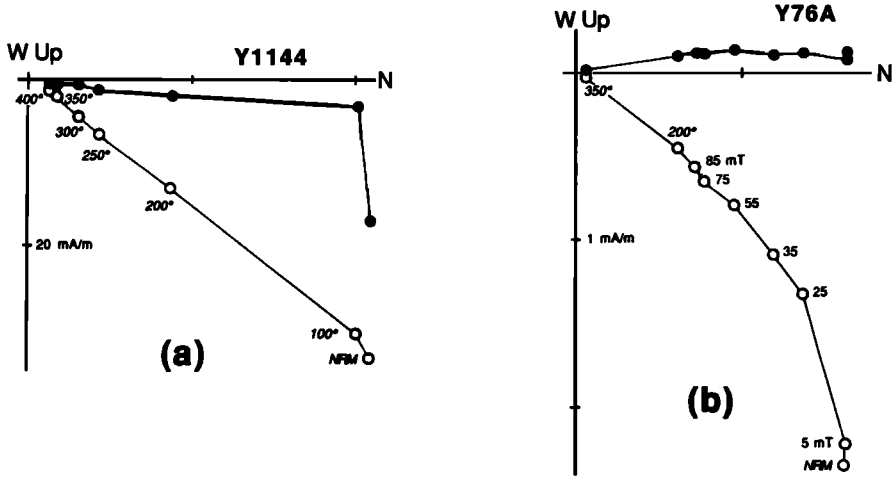


Fig. 4. Orthogonal vector diagrams [Zijderveld, 1967] of representative samples that showed only present-day field geomagnetic overprint directions during thermal (°C) or AF (mT) demagnetization. Plotted points show the end points of the magnetization vector during progressive demagnetization; open (closed) symbols represent projections onto the vertical (horizontal) plane. Magnetization intensities are indicated along the axes. (a) Limestone sample from the Middle Devonian Qingmen Formation of eastern Yunnan. (b) Limestone sample from the Middle Devonian Heyuanzai Formation of western Yunnan.

magnetization (IRM) shows that saturation is not reached in fields up to 1.4 T, indicating that hematite is probably present; whether it is a carrier of the remanence is not certain, given that the maximum unblocking temperatures observed during demagnetization are often around 590°C. Tilt correction slightly improves the clustering of the sample directions (Table 3), but this is not statistically significant.

The Middle Devonian Qingmen Formation. All samples, except five which were unstable, showed only pdf directions (Figure 4a). Upon tilt correction the precision parameter k decreases from 62 to 21, again illustrating that the magnetization is secondary and acquired after the folding. It appears that in this formation, magnetite is the carrier of the remanence.

The Lower Devonian Cuifengshan Formation. AF demagnetization did not succeed in removing a significant portion of the NRM, so all samples were thermally demagnetized. One site yielded characteristic west-northwesterly and shallow directions (e.g., Figure 6d), whereas both sites of this formation also contained pdf directions as well as random or unstable components of magnetization above 250°C.

Summary of Devonian Results from Eastern Yunnan. The four formations sampled contain predominantly pdf directions, but in seven sites characteristic directions were also revealed; these are shown in Figure 7, whereas Table 3 gives the mean (D/I = 279°/+3°) and its statistical parameters based on sample and site-mean directions, as well as the individual formation means. We will return to these results in the discussion section.

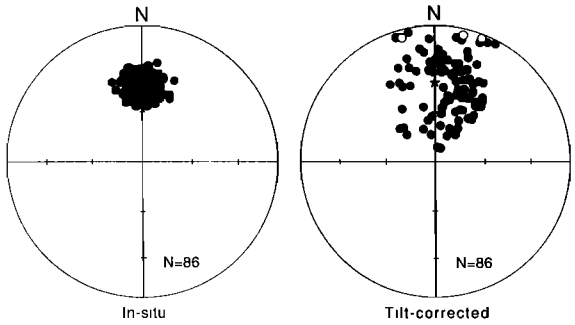


Fig. 5. Equal-area projections of the present-day field overprint directions from eastern and western Yunnan before and after correction for the tilt of the strata. These directions fail the fold test. Full (open) symbols represent directions plotted onto the lower (upper) hemisphere. The star represents today's geomagnetic field direction in Yunnan.

Western Yunnan

Three formations were sampled from the Devonian sequence. The results will again be described per formation from young to old; they are summarized in Table 2.

The Upper Devonian Dazaimen Formation. Because of unfavorable outcrop situations, only one site could be sampled, but all four samples of this site yield a direction of magnetization which is significantly different from the pdf direction and, in view of the results from the other formations of western Yunnan, this component can be called the characteristic magnetization. Figure 8a illustrates the removal of a pdf overprint and the isolation and subsequent elimination of an easterly and upward characteristic direction in in situ coordinates for a sample from the Dazaimen Formation.

The Middle Devonian Heyuanzai Formation. All three sites of this formation revealed characteristic directions, although three samples were unstable and three more had only pdf overprints (Figure 4b). In site 46 a low-blocking temperature pdf overprint dominates the remanence (Figure 8b) and may be

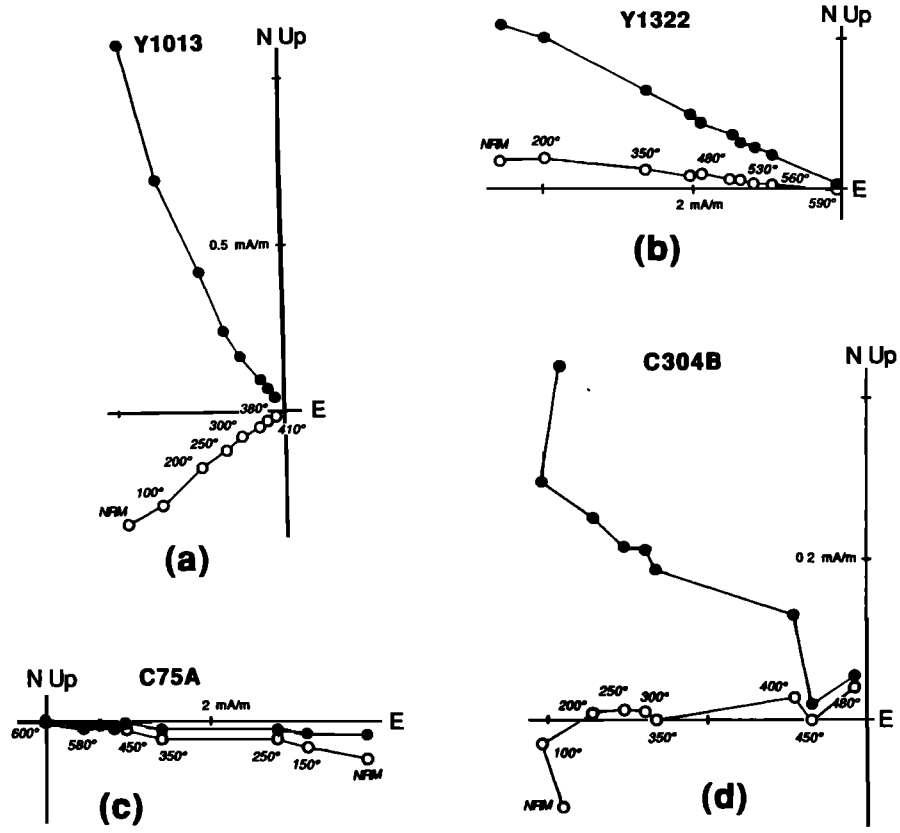


Fig. 6. Zijderveld diagrams (as in Figure 4) of representative samples from eastern Yunnan, showing characteristic directions in in situ coordinates. (a) Limestone sample from the Upper Devonian Zaige Formation. (b) and (c) Sandstone samples from the Middle Devonian Haikou Formation. (d) Sandstone sample from the Lower Devonian Cuifengshan Formation.

carried by goethite. Nevertheless, the characteristic component can be recognized in samples from this site (e.g., Figure 8c) above 300°C, with an in situ shallow and southerly direction. Site 47 also shows multivectorial remanence; after removal of the pdf overprint by about 300°C the characteristic direction (southwesterly and shallow upward in situ) is isolated (Figure 8d). Site 16 has a different structural correction and here the in situ characteristic direction (Figure 8e) is east-northeast and steeply upward, as revealed in thermal demagnetization above 350°C. The different tilt corrections for the sites of the Heyuanzai Formation allow a fold test, which is clearly positive: k increases from 4 to 31 upon unfolding. After correction the characteristic direction is southerly and steeply up, which is similar to the characteristic direction of the Upper Devonian Dazaimen Formation discussed earlier.

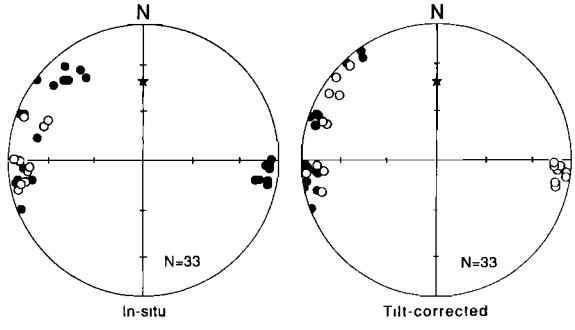


Fig. 7. Equal-area projections, as in Figure 5, of the characteristic directions from eastern Yunnan before and after correction for the tilt of the strata. These directions pass the reversal test (Table 3).

The Lower Devonian Wenquan Formation. Three of the five sites revealed the characteristic direction, whereas the two other sites have only pdf or unstable directions. The characteristic direction of this formation is steeply downward, much steeper than the pdf direction, as can be seen in Figure 8f. After tilt correction the three sites show northerly to northeasterly and steeply down directions (D/I = 21°/+62°), antipodal to the characteristic directions of the overlying Heyuanzai Formation (D/I = 205°/-61°).

Summary of Devonian results from Western Yunnan. Seven sites from the three formations sampled in western Yunnan reveal characteristic directions, which are summarized in Tables 2 and 3 and plotted before and after tectonic

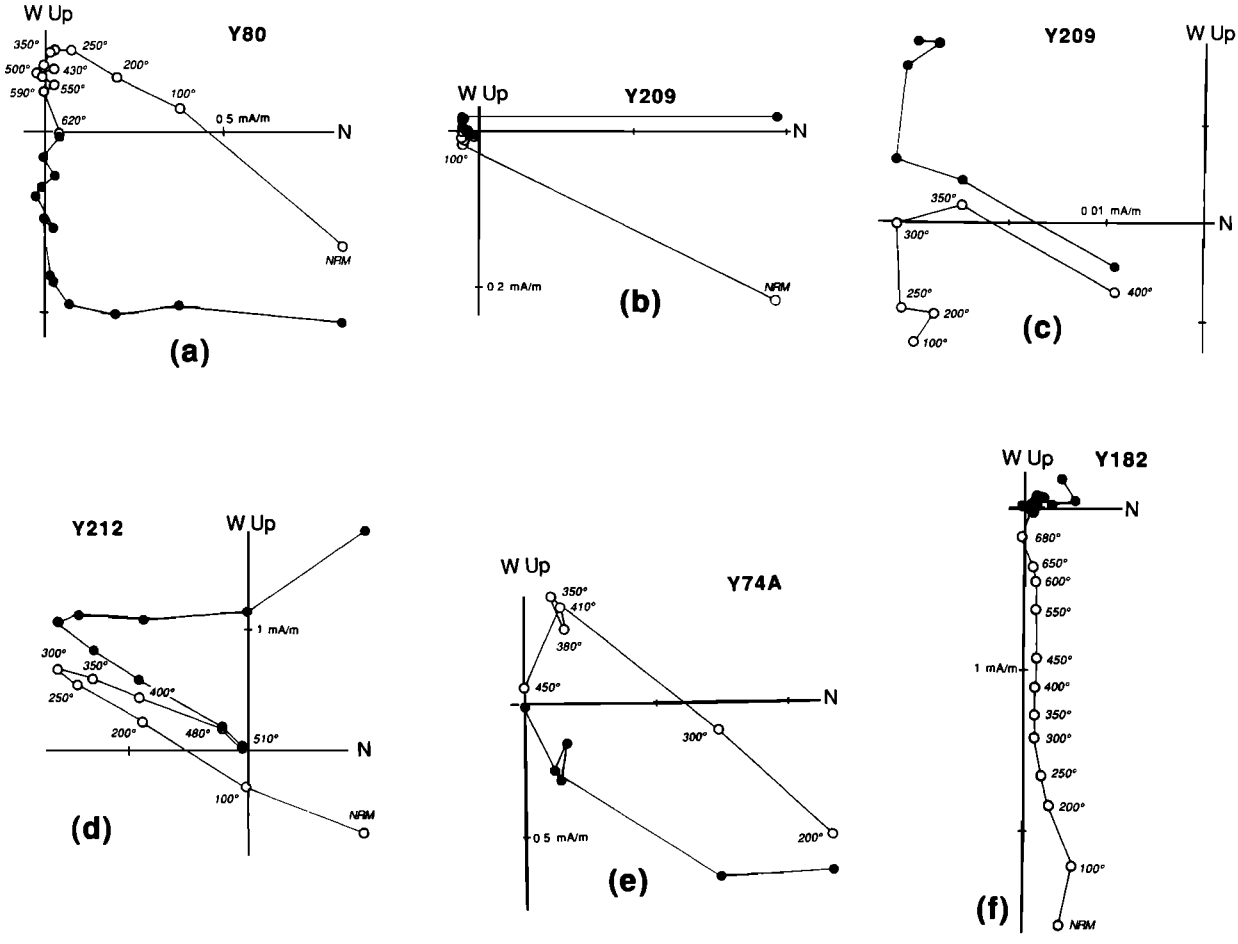


Fig. 8. Zijderveld diagrams (as in Figure 4) of representative samples from western Yunnan showing characteristic directions in in situ coordinates. (a) Limestone sample from the Upper Devonian Dazaimen Formation. (b) - (e) Limestone samples from the Middle Devonian Heyuanzai Formation, with Figure 8c showing enlargement of the higher-temperature portion of the diagram in Figure 8b. (f) Limestone sample from the Lower Devonian Wenquan Formation.

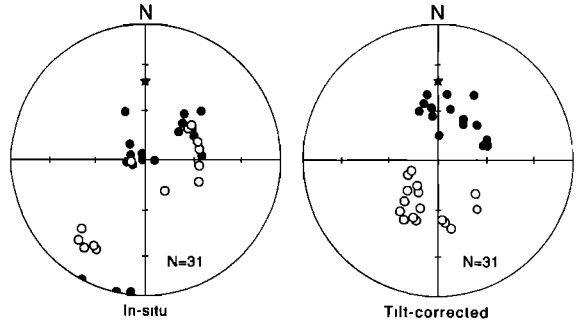


Fig. 9. Equal-area projections, as in Figure 5, of the characteristic directions from western Yunnan before and after correction for the tilt of the strata. These directions pass the reversal test (Table 3) and pass the fold test, with k2/k1 ratios of 4.6 (giving unit weight to N= 31 samples) or 7.1 (N=7 site means) at the 99% confidence level [McElhinny, 1964].

correction in Figure 9. Overall means are given in Table 3, based on site-mean directions as well as sample directions. We prefer the latter, given the variable number of entries per site, because individual sample directions of these sediments, when combined, probably best average out the secular variation. The fold test is clearly positive for these results, indicating that the age of the magnetization is older than that of the folding. In addition, when the results of all three formations are considered together, they pass the reversal test (Table 3). However, folding in the region occurred in Tertiary as well as Early Mesozoic (Carnian-Norian) time [Burrett, 1974], and it is not clear which of these deformation phases primarily affected our sampling area.

DISCUSSION

Despite severe overprinting by a recent geomagnetic field direction, seven sites from eastern Yunnan and seven sites

from western Yunnan have yielded characteristic directions. The latter pass the fold test, and both groups are of dual polarity, passing the reversal test (Table 3). It may seem that a ratio of 64 characteristic directions to a total of 204 samples (from the two areas) is rather low, but in paleomagnetic studies of carbonate rocks elsewhere in China such a low recovery rate has been quite common [e.g., Kent et al., 1987; Zhao, 1987; Fang et al., 1988-89].

Critical to our interpretation in terms of tectonics of southeast Asia is an understanding of the age of magnetization: while the characteristic directions do not resemble results so far obtained for younger periods, the APWPs for the South China Block and for western Yunnan are far from precisely known. Thus a major part of this section will involve a discussion of the possibility that our results represent ancient but secondary magnetizations.

The westerly and shallow characteristic directions from eastern Yunnan are complemented by one site (C 11) with antipodal easterly directions, which supports our interpretation that these directions represent an ancient geomagnetic field direction in which secular variation has been averaged out. While the fold test is not statistically significant, the clustering improves upon tectonic correction; the folding, however, is of Tertiary age [Liang et al., 1984], and even a positive fold test would not constrain the age of magnetization sufficiently to conclude that the magnetizations are primary. On the other hand, the APWP for the South China Block is reasonably well determined for Permian and younger times, and it appears to be very likely that our results are of pre-Permian age. There are also some less well-determined results for the Carboniferous [Lin et al., 1985], and these fall on the APWP for South China between the Late Permian paleopole and the pole (D) from our results (Figure 10). It appears from the APWP of this illustration that a Devonian age for our results would be a reasonable interpretation, given the location of our paleopole on the APWP.

Similar east-west directions have been obtained by Opdyke et al. [1987] for Silurian rocks near Qujing and by Lin et al. [1985] for Lower Cambrian rocks from Hubei, Yunnan, and Zhejiang. Their paleopole positions are also plotted in Figure 10 (S,$\in$l), and the similarity of our Devonian paleopole and those for the Silurian and Early Cambrian rocks is remarkable. Either the Early Paleozoic APWP segment for the South China Block is stationary (and/or looping back on itself) or the older rocks are remagnetized in the Devonian. Further work is necessary to resolve this ambiguity; whatever the outcome, if there are any suspicions about remagnetization, the more likely scenario would involve resetting of the Silurian and Cambrian rocks rather than remagnetization of our Devonian carbonates. We conclude therefore that our paleopole from the Devonian of eastern Yunnan is best interpreted as based on a (near-)primary magnetization. Our interpretation of a Devonian equatorial position of the South China Block is supported by the warm water nature of deposits and faunas of the Yangtze assemblage of that age [Chen et al., 1987; Heckel and Witzke, 1979] (see also the maps of Ziegler [1981]).

Unfortunately, we do not have sufficient sample directions and hence enough resolution to distinguish between Early, Middle and Late Devonian paleopoles. The declinations of the Upper Devonian Zaige Formation are northwesterly whereas

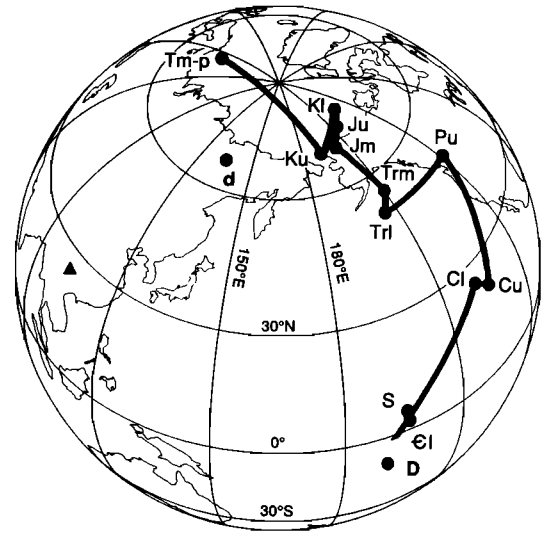


Fig. 10. Apparent Polar Wander Path for the South China Block, including eastern Yunnan. All poles are from Lin et al. [1985] unless otherwise noted. Tm-p, Middle to Late Tertiary; K, Cretaceous; J, Jurassic; Tr, Triassic; P, Permian [Zhao and Coe, 1987]; C, Carboniferous; D, Devonian (this study); S, Silurian [Opdyke et al., 1987]; $\in$, Cambrian; u, Late; m, Middle; l, Early. It is inferred that these poles are north poles. The pole labeled d is our result (south pole?) from western Yunnan, shown for comparison, and the triangle is our sampling area.

those for the older Devonian rocks are more east-west (Table 3). Whether this difference implies a Middle-Late Devonian rotation of the South China Block or secular variation cannot be decided at this time.

The characteristic directions from western Yunnan (assumed to be part of the Shan State Block) contrast sharply with those from the South China Block. The dual-polarity directions are antipodal (Figure 9; Table 3) and pass the fold test (Table 3), but this test only constrains the age of magnetization to pre-Tertiary times. The basic question in the interpretation of the age of this result is whether it could represent a Mesozoic or Late Paleozoic remagnetization; unfortunately, hardly any results for these times are available for a comparison between an APWP for the Shan State Block and our paleopole at 67°N, 134°E (Table 3 and d in Figure 10). The only paleomagnetic result from western Yunnan is labeled "preliminary" and gives a paleolatitude of 34° for the Late Carboniferous [Zhang and Zhang, 1986]. The declination of this result is very different from ours, but the paper does not provide sufficient details to assess the effects of, for instance, the tilt correction. The paleolatitude derived for the sampling area from our result is 42°, and if this represents the northern hemisphere, it is about 17° more northerly than the present-day latitude. Could the Shan State Block have been this far north at any time in the Mesozoic or Late Paleozoic?

Mesozoic and younger paleomagnetic results from eastern Thailand (Indochina Block), the Malay Peninsula, and areas farther away in southeast Asia [Haile, 1981; Chen and Courtillot, 1989] are not necessarily valid for the Shan State

Block, but if they are, then they would support the argument that its Mesozoic and later paleolatitudes were generally equatorial to subtropical, that is, the same or lower than today's. There are thus far no post-Early Triassic results for any parts of Southeast Asia which indicate more northerly paleolatitudes. We tentatively conclude that a plausible interpretation is that the age of the magnetization of our Devonian rocks in western Yunnan is pre-Middle Triassic.

Permian and Early Triassic results from the Malay Peninsula yield paleopoles at 55°N, 164°E and 57°N, 152°E, respectively [McElhinny et al., 1974; Haile, 1981]. For the Shan State Massif these paleopoles would yield extrapolated paleolatitudes of 35° and 41°, respectively. The latter is especially close to the observed paleolatitude of 42° in western Yunnan. Thus we cannot preclude a Permian-Early Triassic age for our results, assuming that the Shan State Block and the Malay Block could have been part of the same (Shan-Thai-Malay) plate.

More exciting, however, is the possibility that our result represents a southerly rather than northerly paleolatitude and that its age is Carboniferous or Devonian. In that case the paleolatitude (42°S) would support a position for the Shan State Block near western Australia and northeastern India (as well as the Lhasa Block of southern Tibet) on the periphery of Gondwana, an area which is located at paleolatitudes between 35°S and 45°S from the Late Devonian to the Late Carboniferous given the combined Gondwana APWP for that time interval [Scotese et al., 1979; Van der Voo, 1988]. Figure 11 shows a Late Devonian paleogeographic map of Gondwana and the equatorial South China Block, with the Shan-Thai-Malay Block positioned according to the 42°S

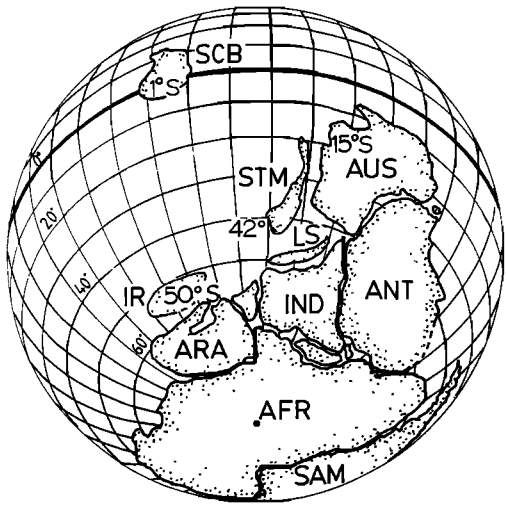


Fig. 11. Paleogeographic map for Late Devonian time [after Van der Voo, 1988; Scotese, 1986]. SCB, South China Block; STM, Shan-Thai-Malay plate; LS, Lhasa Block of southern Tibet; IND, India; AUS, Australia; ANT, Antarctica; AFR, Africa; IR, Iran; ARA, Arabia; SAM, South America. Longitudes are arbitrary; the SCB, for instance, may be displaced east or westward with respect to Gondwana. Paleomagnetically determined paleolatitudes are indicated for SCB, STM (this study), cratonic Australia [Van der Voo, 1988] and Iran [Wensink et al., 1978].

paleolatitude option. There is considerable geological evidence for a cold water regime during the Late Paleozoic in the rocks of the Shan State Massif, that is, west of the Lancang River fault Zone [Cao, 1986; Chen et al., 1987], as well as further south in Thailand and the Malay Peninsula (Ridd, [1971], and Waterhouse, quoted in Stauffer [1983]). Permian fossils, such as Glossopteris, for this time show typical Gondwana affinities [Ridd, 1971; Stauffer, 1983] (see also Crawford [1973]), although the Malayan paleomagnetic data show Permian paleolatitudes that are somewhat too low for immediate Gondwana proximity; several authors [e.g., Burrett, 1974; Burrett and Stait, 1988] have, moreover, been ambivalent about an immediate proximity to Gondwana in the Permian on other grounds. A possible explanation could be that these pieces of Southeast Asia were disjoint in the Permian and Early Mesozoic and/or that some of them had separated from Gondwana to travel northward in the Tethys [e.g., Scotese et al., 1979, figures 35 and 39].

Problematic with the interpretation that the Shan State Block was adjacent to western Australia is that the Early to Middle Devonian APWP segment of Gondwana would predict lower paleolatitudes for this location than the ones predicted for the Late Devonian [Van der Voo, 1988]; yet our results are based on Lower, Middle, and Upper Devonian rocks from the Shan State Massif, with the individual formation-mean inclinations all being very similar. Despite the presence of normal and reversed polarities, this would then imply that the magnetizations are not primary but all acquired during the Late Devonian or Carboniferous.

Summarizing our conclusions about the age of the magnetizations in western Yunnan, we conclude that a Late Paleozoic age, bracketed between Devonian and Middle Triassic, best fits the currently available paleogeographic and paleomagnetic data, assuming that the Shan State Massif, western Thailand, and the Malay Peninsula were all part of the same (Shan-Thai-Malay) Block. Noting that our results are of dual polarity, one could argue, moreover, that the magnetizations are not of Kiaman age; this would imply either a Devonian-Early Carboniferous or a latest Permian-Early Triassic age of the magnetization. What seems clear, however, regardless of the age, is that the steeply inclined directions observed in western Yunnan are significantly different from any Devonian or younger directions observed in the South China Block. Thus our paleomagnetic observations lead us to conclude that a suture must be placed between the Shan State Massif and the Yangtze Paraplatform; the Lancang River or Red River fault zones are likely candidates for this suture (whether strike-slip or convergent-oblique).

It must be noted that the Gondwana affinity of the Shan-Thai-Malay fauna and a Late Paleozoic magnetization age, as well as the arguments for colder water deposition [Stauffer, 1983] define the southerly and up direction as one of normal polarity (southern hemisphere paleolatitude). This implies that western Yunnan rotated 160° with respect to the present meridian since the magnetization was acquired. By inference, a similar rotation can be inferred for the Late Paleozoic-Early Triassic results from the Malay Peninsula, as well as for the Ordovician paleomagnetic result from the Setul Formation of the Malay Peninsula [Haile, 1981]. Such an "upside down" orientation adjacent to Gondwana is not illogical: Audley-

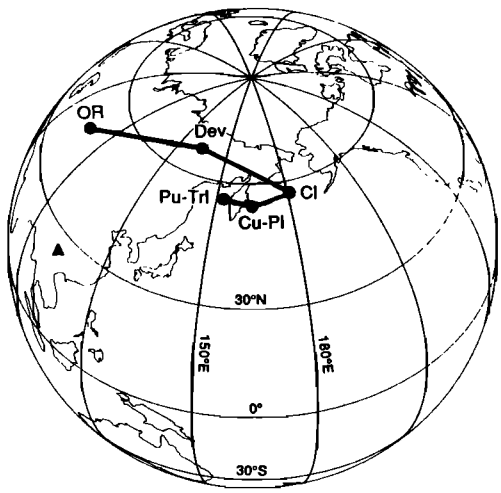


Fig. 12. Tentative Apparent Polar Wander Path (as in Figure 10) for the Shan-Thai-Malay plate, including western Yunnan. Paleopoles are from the review by Haile [1981] with the addition of our result from western Yunnan (Dev). OR, Ordovician; other abbreviations as in Figure 10. It is inferred that most of these poles are south poles (see text). The solid triangle is our sampling area.

Charles [1983] and Scotese [1986] place the Shan-Thai-Malay Block in their Paleozoic reconstructions precisely in this orientation, very similar to that shown in Figure 11, and Stauffer [1983] has argued that it was the present-day west side of the Malay Peninsula which was adjacent to a larger continental mass (such as Australia in Figure 11). Last, Burrett and Stait [1988] concluded that early Paleozoic faunal similarities between the Shan-Thai Block and Australia were so striking (e.g., a generic Simpson Index for nautiloids of 0.92) that the two must have been contiguous. In contrast, the similarities between the Shan-Thai and the South China Blocks were very small (generic Simpson Index of 0.08).

A tentative APWP for the Shan-Thai-Malay Block is constructed in Figure 12, on the basis of the poles listed in the review of Haile [1981], with the addition of our new pole (Dev). All these results are inferred to be south poles. A comparison with the APWP for the South China Block (Figure 10), likely to be based on north poles, illustrates the rather different tectonic drift history of these two blocks and a large relative rotation. We note that in the southeast Asian extrusion model [e.g., Peltzer and Tapponnier, 1986], about 60° of clockwise rotation of the Shan-Thai-Malay Block is implied. This goes a long way toward explaining the large rotation inferred from this study.

CONCLUSIONS

The Devonian rocks from eastern Yunnan carry a (near-) primary magnetization which indicates an equatorial paleoposition for the South China Block. The similarity between our Devonian paleopole and those for Silurian and Early Cambrian rocks from the South China Block warrants further investigation to check whether the latter represent remagnetizations.

The Devonian from western Yunnan yields a paleomagnetic result with fairly high paleolatitude (42°), which contrasts with the lower Phanerozoic paleolatitudes of the South China Block, indicating that eastern and western Yunnan were spatially separated. The age of the magnetization of the western Yunnan rocks is interpreted to be Devonian (to possibly Early Triassic), and if this is true, it confirms a paleolocation of the Shan State Massif in the periphery of Gondwana. We tentatively propose a location for the Shan-Thai-Malay plate adjacent to northwestern Australia, northeastern India, and southern Tibet, with the present-day northern end facing the south pole. This location is in agreement with previous reconstructions based on other, nonpaleomagnetic arguments.

Assuming the Shan State Massif, Thailand, and the Malay Peninsula were all part of the same (Shan-Thai-Malay) plate throughout their history, this plate began to travel northward away from Gondwana in the Permocarboniferous or, at the latest, after the Early Triassic. It may have joined the South China Block by the Late Triassic [Achache and Courtillot, 1985], which could explain the orogenic phase of deformation of that age in the zone between the Shan State Massif and the South China Block, but there is insufficient paleomagnetic evidence to confirm this. A large cumulative rotation of about 160° of the Shan-Thai-Malay plate may have happened in stages, first as it drifted northward and later because of the extrusion caused by the India-Asia collision process.

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REFERENCES

Achache, J., and V. Courtillot, A preliminary Upper Triassic paleomagnetic pole for the Khorat plateau (Thailand): Consequences for the accretion of Indochina against Eurasia, *Earth Planet. Sci. Lett.*, *73*, 147-157, 1985.

Argand, E., La tectonique de l'Asie, *Comptes Rendus Congres Geol. Intern., 13th*, pp. 171-372, Imprimerie H. Vaillant-Carmanne, Liege, 1924.

Audley-Charles, M. G., Reconstruction of Gondwanaland, *Nature*, *306*, 48-50, 1983.

Burrett, C. F., Plate tectonics and the fusion of Asia, *Earth Planet. Sci. Lett.*, *21*, 181-189, 1974.

Burrett, C. F. and B. Stait, Early Paleozoic biogeography and paleomagnetism of greater Gondwana (abstract), in *Palaeozoic Biogeography and Palaeogeography*

Symposium, Abstr. Vol., edited by C. R. Scotese and W. S. Mckerrow, p. 9, Oxford Univ., Oxford, U. K., 1988.

Cao, R. G., Discovery of Late Carboniferous glacio-marine deposits in western Yunnan (in Chinese with English abstract), *Geol. Rev.*, *32*, 236-242, 1986.

Chan, L. S., C. Y. Wang, and X. Y. Wu, Paleomagnetic results from some Permian-Triassic rocks from southwestern China, *Geophys. Res. Lett.*, *11*, 1157-1160, 1984.

Chen, B. W., K. Y. Wang, W. X. Liu, Z. J. Chai, Q. W. Zhang, X. J. Peng, Y. Z. Qiu, and T. Z. Zheng, Tectonics of the Nu River, Lancang River and Jinsha River area (in Chinese), 195 pp.,Geological Publishing House, Beijing, China, 1987.

Crawford, A. R., A displaced Tibetan massif as a possible source of some Malayan rocks, *Geol. Mag.*, *109*, 483-489, 1973.

Fang, W., R. Van der Voo, and Q. Z. Liang, Reconnaissance magnetostratigraphy of the Precambrian-Cambrian boundary section at Meishucun, China, *Cuad. Geol. Iber.*, *12*, 205-222, 1988-89.

Haile, N. S., Paleomagnetism of southeast and east Asia, in *Paleoreconstruction of the Continents, Geodyn. Ser.*, vol. 2, edited by M. W. McElhinny and D. A. Valencio, pp. 129-135, AGU, Washington, D.C., 1981.

Heckel, P. H., and B. J. Witzke, Devonian world palaeogeography determined from distribution of carbonates and related lithic palaeoclimatic indicators, in *Special Papers in Palaeontology*, vol. 23, pp. 99-123, Palaeontological Association, University Press, Oxford, England, 1979.

Huang, K. N., N. D. Opdyke, G. Z. Xu, and R. L. Tang, Further paleomagnetic results from the Permian Emeishan Basalt in SW China, *Kexue Tongbao* (Engl. Transl.), *31*, 1195-1201, 1986.

Kent, D. V., G. Xu, K. Huang, W. Y. Zhang, and N. D. Opdyke, Paleomagnetism of Upper Cretaceous rocks from South China, *Earth Planet. Sci. Lett.*, *79*, 184-189, 1986.

Kent, D. V., X. S. Zeng, W. Y. Zhang, and N. D. Opdyke, Widespread late Mesozoic to Recent remagnetization of Paleozoic and Lower Triassic sedimentary rocks from South China, *Tectonophysics*, *139*, 133-143, 1987.

Kirschvink, J. L., The least-squares line and plane and the analysis of paleomagnetic data, Geophys. *J. R. Astron. Soc.*, *62*, 699-718, 1980.

Li, C. Y., Q. Wang, X. Y. Liu, and Y. Q. Tang, *Explanatory notes to the tectonic map of Asia*, 49 pp., Cartographic Publishing House of China, Beijing, China, 1982.

Liang, Q. Z., et al., *Report on the regional geological investigation in the Kunming District*, Yunnan Geological Bureau, Kunming, China, 1984.

Lin, B. Y., *The Silurian System of China*, 16 pp., Inst. of Geol., Chinese Acad. Geol. Sci., Beijing, China, 1980.

Lin, J. L., M. D. Fuller, and W. Y. Zhang, Preliminary Phanerozoic polar wander paths for the North and South China Blocks, *Nature*, *313*, 444-449, 1985.

Liu, C., X. X. Zhao, R. S. Coe, and Q. Z. Liang, Paleomagnetic evidence: The anticlockwise rotation of the Qujing area during the Paleozoic (in Chinese), *Sci. Geol. Sin.*, *3*, 232-235, 1986.

McElhinny, M. W., Statistical significance of the fold test in palaeomagnetism, Geophys. *J. R. Astron. Soc.*, *8*, 338-340, 1964.

McElhinny, M. W., N. S. Haile, and A. R. Crawford, Paleomagnetic evidence shows Malay Peninsula was not part of Gondwanaland, *Nature*, *252*, 641-645, 1974.

McElhinny, M. W., B. J. J. Embleton, X. H. Ma, and X. K. Zhang, Fragmentation of Asia in the Permian, *Nature*, *293*, 312-316, 1981.

Miller, J. D., and D. V. Kent, Regional trends in the timing of Alleghenian remagnetization in the Appalachians, *Geology*, *16*, 588-591, 1988.

Mitchell, A. H. G., Phanerozoic plate boundaries in mainland SE Asia, the Himalayas and Tibet, *J. Geol. Soc., London*, *138*, 109-122, 1981.

Opdyke, N. D., K. Huang, G. Xu, W. Y. Zhang, and D. V. Kent, Paleomagnetic results from the Triassic of the Yangtze Platform, *J. Geophys. Res.*, *91*, 9553-9568, 1986.

Opdyke, N. D., K. Huang, G. Xu, W.Y. Zhang, and D. V. Kent, Paleomagnetic results from the Silurian of the Yangtze Paraplatform, *Tectonophysics*, *139*, 123-132, 1987.

Peltzer, G., and P. Tapponnier, Formation and evolution of strike-slip faults, rifts, and basins during the India-Asia collision: An experimental approach, *J. Geophys. Res.*, *93*, 15,085-15,117, 1988.

Ridd, M. F., Southeast Asia as a part of Gondwanaland, *Nature*, *234*, 531-533, 1971.

Scotese, C. R., Phanerozoic reconstructions: A new look at the assembly of Asia, *Tech. Rep.*, *66*, 54 pp., Institute for Geophysics, Univ. of Texas, Austin, 1986.

Scotese, C. R., R. K. Bambach, C. Barton, R. Van der Voo, and A. M. Ziegler, Paleozoic base maps, *J. Geol.*, *87*, 217-277, 1979.

Stauffer, P. H., Unraveling the mosaic of Paleozoic crustal blocks in southeast Asia, *Geol. Rundsch.*, *72*, 1061-1080, 1983.

Tapponnier, P., G. Peltzer and R. Armijo, On the mechanics of the collision between India and Asia, in *Collision Tectonics, Spec. Publ.*, *19*, edited by M. P. Coward and A. C. Ries, pp. 115-157, Geological Society of London, 1986.

Van der Voo, R., Paleozoic paleogeography of North America, Gondwana, and intervening displaced terranes: Comparisons of paleomagnetism with paleoclimatology and biogeographical patterns, *Geol. Soc. Am. Bull.*, *100*, 311-324, 1988.

Van der Voo, R., Paleomagnetism of North America; the craton, its margins and the Appalachian Belt, in *Geophysical Framework of the Continental United States, Mem. 172*, edited by L. C. Pakiser and W. D. Mooney, Geological Society of America, Boulder, Colo., in press, 1989.

Wensink, H., J. D. A. Zijderveld, and J. C. Varekamp, Paleomagnetism and ore mineralogy of some basalts of the Geirud Formation of Late Devonian - Early Carboniferous age from the southern Alborz, Iran, *Earth Planet. Sci. Lett.*, *41*, 441-450, 1978.

Whittington, H. B., and C. P. Hughes, Ordovician geography

and faunal provinces, *Philos. Trans. R. Soc. London, 263*, 235-278, 1972.

Yan, C., and V. Courtillot, Widespread Cenozoic (?) remagnetization in Thailand and its implications for the India-Asia collision, *Earth Planet. Sci. Lett.,93*, 113-122, 1989.

Yang, T., Y. Cheng, and H. Wang, *The Geology of China*, 363 pp., Clarendon, Oxford, U. K., 1986.

Zhang, Z. K., and J. X. Zhang, Paleomagnetic research on the Upper Carboniferous basalt in the Baoshan block, Yunnan, and the tectonic belonging of the block (in Chinese with English abstract), *Geol. Inst. Bull. Chin. Acad. of Geol. Sci., 15*, 183-189, 1986.

Zhao, X. X., A paleomagnetic study of Phanerozoic rock units from eastern China, Ph. D. thesis, 401 pp., Univ. of Calif. Santa Cruz, 1987.

Zhao, X. X., and R. S. Coe, Paleomagnetic constraints on the collision and rotation of North and South China, *Nature, 327*, 141-144, 1987.

Ziegler, A. M., Paleozoic paleogeography, in *Paleoreconstruction of the Continents, Geodyn. Ser.*, vol. 2, edited by M. W. McElhinny and D. A. Valencio, pp. 31-37, AGU, Washington, D.C., 1981.

Zijderveld, J. D. A., AC demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism*, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254-286, Elsevier, Amsterdam, 1967.

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